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Department of Information Technology (Bsc.IT)
Traffic Light Management System
DS Project

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CODE:

```
signals = []
```

```
def find(id):  
    for s in signals:  
        if s["id"] == id:  
            return s  
    return None
```

```
def add_signal():  
    id = int(input("Enter Signal ID: "))  
    light = input("Enter Light (RED/YELLOW/GREEN): ").upper()
```

```
    signals.append({  
        "id": id,  
        "light": light,  
        "vehicles": [],  
        "history": [light]  
    })
```

```
    print("Signal added.")
```

```
def display():  
    print("\nID---Light---Vehicles---")  
    for s in signals:  
        print(f'{s["id"]} {s["light"]} {len(s["vehicles"])}')"
```

```
def search():  
    id = int(input("Enter ID: "))  
    s = find(id)
```

```
    if not s:
```

```

        print("Not found.")
        return

    print(f"ID:{s['id']} Light:{s['light']} Vehicles:{len(s['vehicles'])}")

def add_vehicle():
    s = find(int(input("Enter Signal ID: ")))

    if not s:
        print("Not found.")
        return

    v = input("Enter Vehicle No: ")
    s["vehicles"].append(v)
    print("Vehicle added.")

def remove_vehicle():
    s = find(int(input("Enter Signal ID: ")))

    if not s:
        print("Not found.")
        return

    if s["vehicles"]:
        v = s["vehicles"].pop(0)
        print("Vehicle removed:", v)
    else:
        print("No vehicles waiting.")

def show_waiting():
    s = find(int(input("Enter Signal ID: ")))

    if not s:
        print("Not found.")
        return

    print("Waiting Vehicles:", s["vehicles"])

def change_light():
    s = find(int(input("Enter Signal ID: ")))

    if not s:
        print("Not found.")
        return

    light = input("New Light: ").upper()
    s["light"] = light
    s["history"].append(light)

```

```

print("Light changed.")

def show_history():
    s = find(int(input("Enter Signal ID: ")))

    if not s:
        print("Not found.")
        return

    print("History:", " " -> ".join(s["history"]), "-> END")

def sort_by_traffic():
    n = len(signals)

    for i in range(n):
        for j in range(0, n - i - 1):
            if len(signals[j]["vehicles"]) < len(signals[j + 1]["vehicles"]):
                signals[j], signals[j + 1] = signals[j + 1], signals[j]

    print("Sorted by traffic (highest first).")
    display()

def menu():
    while True:
        print("\n=====")
        print("1. Add Traffic Signal")
        print("2. Display Traffic Signals")
        print("3. Search Traffic Signal")
        print("4. Add Vehicle")
        print("5. Remove Vehicle")
        print("6. Show Waiting Vehicles")
        print("7. Change Traffic Light")
        print("8. Show Signal History")
        print("9. Sort Signals by Traffic")
        print("10. Exit")

        choice = input("Enter your choice: ")

        actions = {
            "1": add_signal,
            "2": display,
            "3": search,
            "4": add_vehicle,
            "5": remove_vehicle,
            "6": show_waiting,
            "7": change_light,
            "8": show_history,
            "9": sort_by_traffic

```

```
}

if choice == "10":
    print("THANK YOU")
    break
elif choice in actions:
    actions[choice]()
else:
    print("Invalid choice.")

if __name__ == "__main__":
    menu()
```

Output:

```
=====
1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit
Enter your choice: 1
Enter Signal ID: 101
Enter Light (RED/YELLOW/GREEN): RED
Signal added.
```

```
=====
1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit
Enter your choice: 1
Enter Signal ID: 102
Enter Light (RED/YELLOW/GREEN): YELLOW
Signal added.
```

2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit

Enter your choice: 2

--ID--Light--Vehicles--

101 RED 0
102 YELLOW 0
103 GREEN 0

=====

1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit

Enter your choice: 3

Enter ID: 103

ID:103 Light:GREEN Vehicles:0

=====

1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle

=====

1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit

Enter your choice: 4

Enter Signal ID: 101

Enter Vehicle No: 000

Vehicle added.

=====

1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit

Enter your choice: 4

Enter Signal ID: 102

Enter Vehicle No: 0000

Vehicle added.

=====

```
=====
1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit
Enter your choice: 4
Enter Signal ID: 103
Enter Vehicle No: 00000
Vehicle added.
```

```
=====
1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit
Enter your choice: 2

--ID--Light--Vehicles--
101  RED   1
102  YELLOW 1
103  GREEN 1
```

- ```

1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit
```

Enter your choice: 3

Enter ID: 103

ID:103 Light:GREEN Vehicles:1

- ```
=====
1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit
```

Enter your choice: 5

Enter Signal ID: 102

Enter Vehicle No: 0000

Vehicle removed.

=====

1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit

Enter your choice: 6

Enter Signal ID: 102

Waiting vehicles: 0

=====

1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit

Enter your choice: 6

Enter Signal ID: 103

Waiting vehicles: 1

00000

=====

1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit

Enter your choice: 7

Enter Signal ID: 103

New Light: RED

Light changed.

```
1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit
Enter your choice: 8
Enter Signal ID: 103
History: GREEN -> RED -> END
```

```
=====
1. Add Traffic Signal
2. Display Traffic Signals
3. Search Traffic Signal
4. Add Vehicle
5. Remove Vehicle
6. Display Waiting Vehicles
7. Change Traffic Light
8. Show Signal History
9. Sort Signals by Traffic
10. Exit
Enter your choice: 9
Sorted by traffic (highest first).

--ID--Light--Vehicles--
101  RED   1
103  RED   1
102  YELLOW 0
```

```
=====
```

-
1. Add Traffic Signal
 2. Display Traffic Signals
 3. Search Traffic Signal
 4. Add Vehicle
 5. Remove Vehicle
 6. Display Waiting Vehicles
 7. Change Traffic Light
 8. Show Signal History
 9. Sort Signals by Traffic
 10. Exit

Enter your choice: 10

Exiting...